

# assign\_functional\_type\_clusters.R

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*Assign Functional-Type Clusters*

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## Description

Assigns taxa into functional types (FTs) using a pre-computed Gower dissimilarity matrix, a pre-fitted hierarchical clustering object, and a predetermined number of groups ‘functional\_type\_group\_count’. The number of groups is expected to have been chosen upstream via ‘select\_functional\_type\_group\_count()’.

## Usage

```
assign_functional_type_clusters(  
  data_trait_table,  
  trait_dissimilarity,  
  hierarchical_clustering,  
  functional_type_group_count,  
  taxon_column = "taxon_name",  
  verbose = TRUE  
)
```

## Arguments

**data\_trait\_table**  
A data frame with one row per taxon. Must contain a character column (identified by ‘taxon\_column’) holding taxon names. Must have at least 4 rows and ‘nrow(data\_trait\_table)’ must be strictly greater than ‘functional\_type\_group\_count’.

**trait\_dissimilarity**  
A ‘dist’ object produced by ‘compute\_trait\_dissimilarity()’. Must inherit class “dist”. All distance values must be finite.

**hierarchical\_clustering**  
An ‘hclust’ object produced by ‘fit\_hierarchical\_clustering()’. Must inherit class “hclust”.

|                                          |                                                                                                                                                                        |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>functional_type_group_count</code> | A single positive integer giving the number of functional-type groups to cut the dendrogram at. Must be at least 2 and less than <code>nrow(data_trait_table)</code> . |
| <code>taxon_column</code>                | A single character string naming the column in <code>'data_trait_table'</code> that contains taxon names. Default: <code>"taxon_name"</code> .                         |
| <code>verbose</code>                     | A single logical. If <code>'TRUE'</code> (default), prints the chosen <code>'functional_type_group_count'</code> via <code>'cli::cli_inform()'</code> .                |

## Details

**\*\*Algorithm\*\*:**

1. Cut the dendrogram at `'functional_type_group_count'` via `'stats::cutree()'`.
2. Compute per-taxon silhouette widths via `'cluster::silhouette()'`.
3. Return a tibble of taxon names, cluster assignments, and silhouette widths.

Distance computation (Inf -> NA, daisy, NaN -> 1.0) and hierarchical clustering are handled upstream by `'compute_trait_dissimilarity()'` and `'fit_hierarchical_clustering()'` respectively. FT-groups selection is handled upstream by `'select_functional_type_group_count()'`. If `'nrow(data_trait_table) < 4'` or `'functional_type_group_count >= nrow(data_trait_table)'`, the function aborts with an informative error.

## Value

A tibble with columns:

**taxon\_name** Taxon names (same values as `'data_trait_table[[taxon_column]]'`). Column name is always `"taxon_name"` regardless of `'taxon_column'`.

**functional\_type** Integer label (1..`'functional_type_group_count'`) giving the functional-type cluster assignment for each taxon.

**silhouette\_width** Per-taxon silhouette width for the chosen `'functional_type_group_count'` solution.

## See Also

`[compute_trait_dissimilarity()]`, `[fit_hierarchical_clustering()]`, `[select_functional_type_group_count()]`

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